

2023

JOURNEYMAN

OPEN

BOOK

EXAM #1



25 QUESTIONS

TIME LIMIT - 1 HOUR

TIME SPENT **MINUTES**

SCORE **%**

JOURNEYMAN OPEN BOOK EXAM #1**25 Questions One Hour Time Limit**

1. Each multiwire branch circuit shall be provided with a means that will simultaneously disconnect all ____.

(a) ungrounded (b) neutral (c) equipment grounding (d) grounded

2. The connection between the grounded circuit conductor and the equipment grounding conductor, or the supply-side bonding jumper, or both at the service is recognized as the ____.

(a) neutral conductor (b) equipment bonding jumper
(c) main bonding jumper (d) grounding electrode conductor

3. When calculating floor area for branch circuit load calculations, the floor area is measured from the ____ dimensions of the building, dwelling unit, or area involved.

(a) drawing (b) scale (c) outside (d) inside

4. NM cable shall be permitted to be laid in notches of wooden studs where the NM cable at those points is protected by a steel plate at least ____ thick to cover the area of the wiring.

(a) 1/16" (b) 3/32" (c) 1/8" (d) 3/16"

5. A motor control center in an equipment room requires GFCI protected 125-volt, single-phase, 15 or 20 amp rated receptacle outlet within ____ feet.

(a) 25 (b) 30 (c) 45 (d) 50

6. Alternating current snap switches shall be permitted for control of inductive loads not exceeding ____ of the ampere rating of the switch at the applied voltage.

(a) 70% (b) 60% (c) 50% (d) 33%

7. Other than the required interconnections and control wiring, only those conductors that are intended for termination in a/an ____ section of a switchboard or switchgear shall be located in that section.

(a) isolated (b) horizontal (c) vertical (d) dual

8. Plate electrodes shall be installed not less than ____ below the surface of earth.

(a) 2 feet (b) 2 1/2 feet (c) 3 feet (d) 4 feet

9. Each grounded conductor shall terminate within the panelboard in a/an ____ terminal that is not also used for another conductor.

- (a) single (b) multiwire (c) dual (d) individual

10. In dwelling units, GFCI protection is required for all 15 and 20 ampere, 125 volt receptacles are installed within ____ feet of the outside edge of a bathtub or shower stall.

- (a) 4 (b) 6 (c) 8 (d) 12

11. When normally enclosed live parts are exposed for inspection or servicing, the working space, if in a passageway or general open space, shall be suitably ____.

- (a) isolated (b) locked (c) guarded (d) insulated

12. All 125-volt through 250-volt receptacles supplied by single-phase branch circuits rated 150 volts or less to ground in kitchens or areas with a/an ____ and permanent provisions for either food preparation or cooking shall have GFCI protection for personnel.

- (a) sink (b) oven (c) range (d) fryer

13. As it relates to load calculations, calculations shall be permitted to be rounded to the nearest whole ampere, with decimal fractions smaller than ____ dropped.

- (a) 0.5 (b) 0.05 (c) 0.6 (d) 0.8

14. GFCI protection shall be provided for lighting outlets not exceeding ____ volts installed in crawl spaces.

- (a) 115 (b) 120 (c) 150 (d) 300

15. Each section of equipment that requires rear or side access to make field connections shall be so marked by the manufacturer on the ____.

- (a) front (b) right side (c) left side (d) rear

16. In other than one and two family dwellings, a receptacle outlet for indoor service equipment shall be located within ____ of the service equipment.

- (a) 25 feet (b) 50 feet (c) 75 feet (d) the same room or area

17. Insulated conductors used inside switchgear or switchboards are required to be ____.

- (a) suitable (b) labeled (c) acceptable (d) listed

18. GFCI protection shall be provided for ____ receptacles.

- (a) equipment requiring servicing (b) permanently installed fire alarm
(c) permanently installed burglar alarm (d) industrial laboratories

19. All lamps for general illumination in temporary wiring installations shall be protected from accidental contact or breakage by a suitable ____ or lampholder with a guard.

- (a) screen (b) luminaire (c) barrier (d) guard

20. In addition to the number of 120 volt, single-phase branch circuits required in other areas of a dwelling, at least one additional ____ ampere branch circuit shall be provided to supply the laundry receptacle outlet in the laundry room. The circuit shall have no other outlets.

- (a) 15 (b) 20 (c) 25 (d) 30

21. Disregarding demand factors, the calculated lighting load for a 5,000 sq.ft. office building is ____ volt-amperes.

- (a) 16,500 (b) 15,400 (c) 8,000 (d) 6,500

22. Appliance receptacle outlets installed in a dwelling unit for specific appliances, such as laundry equipment, shall be installed within ____ of the intended location of the appliance.

- (a) 24" (b) 36" (c) 48" (d) 72"

23. Decorative lighting and similar accessories used for holiday lighting and similar purposes shall be listed and ____.

- (a) marked (b) approved (c) labeled (d) stamped

24. Outlets supplying all pool motors on branch circuits rated 150 volts or less to ground and ____ amperes or less, single or 3 phase, shall be provided with Class A GFCI protection.

- (a) 60 (b) 50 (c) 40 (d) 30

25. Disconnecting means shall be located within sight from, and ____ from air-conditioning or refrigerating equipment.

- (a) serviceable (b) isolated (c) accessible (d) readily accessible

